

GRADIENT INDEX GLASS AND APPLICATIONS

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1. ABSTRACT

The paper reflects on use of gradient index glass in the twenty first century. It's use as a new age material in fields such as medicine, communication, optics as coupling device for fast transmission in fiber optics and endoscopy for modern medicine with its Complexities in manufacturing is highlighted in this paper.

Keyword: Optics, Bending of Light, Refractive Index, Graded-Index Optics, Glass

2. INTRODUCTION

Conventionally Light Rays travels in a straight line with the speed of light. When the same light passes through a GRIN (Gradient Index) optical component it shows phenomenon of bending of light in a peculiar way. It is due to the lenses continuous variation of refractive index such as of a spatial coordinate.

The subject of gradient-index optics dates from the 1850s. However, only in the last 10 years has it been possible to design lenses, manufacture materials, measure the properties, and fabricate finished lens elements.[1] Nippon Sheet Glass Company (Sagamihara, Japan) and NEC¹ first developed Gradient index optical fiber in 1968. [2]. Since then a lot of process have been published to manufacture gradient index glass exploring chemical vapor deposition to the basic as of Ion-Exchange [3].

3. MATERIAL AND METHODS

3.1 Types of Gradient index Lens

There are mainly 4 types of GRIN Lenses made i.e. Axial which has a varying refractive index along a optical plane, Radial in which there is varying index profile from the optical axis to the periphery along the transverse direction, Spherical which has a varying concentric circles radiating from a point with varying refractive index and Periodic.[4]. A curved trajectory is followed in a GRIN glass. Commercially available gradient optics use small optical apertures less than 3 millimeters. This is done in order to make instruments comfortable for the use on patients in case of medical use cases.

3.2 Fabrication of Gradient Index Glass

This is done in order to relieve the stress after which the process gives a parabolic index profiled glass with a diameter of few millimeters. Also there are other few ways of manufacturing of gradient index glass

3.2.1 Ion Exchange [1]

GRIN Glasses are mostly manufactured by using conventional ion-exchange. A highly polarized monovalent ion like Cs^+ or Tl^+ is immersed in a molten salt bath containing K^+ and Na^+ , at a temperature above T_g [5]. Different range is gradient profile is found to be produced

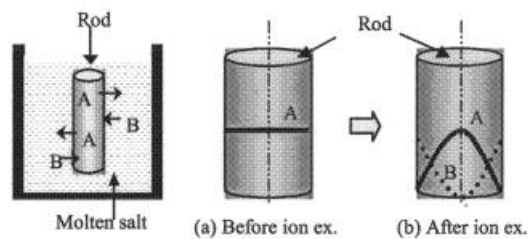


Fig 1 Schematic Diagram of Ion Exchange [6]

3.2.2 Molecular Stuffing [6]

A type of glass is selected which after phase transformation and a dissolving acid bath gives of a glass sponge. This material is etched in sand bath to fill the voids by molecular stuffing. The glass is then recondensed by heating.

3.2.3 Neutron Irradiation [7]

A neutron flux is passed through the glass to alter properties of the glass. One Downside it also causes undesirable changes including radioactivity, production of Color Centers in the glass.

After the glasses are produced it can be then used in different instruments accordingly.

4. APPLICATION IN THE NEW WORLD

Gradient index glass – A new age material has a vast applications in the real world
The main applications of GRIN Lenses are

- Fiber collimators, where a GRIN lens may be fused to a fiber end
- Fiber-to-fiber coupling
- Mode field adapters
- focusing applications, e.g. optical data storage
- Monolithic solid-state lasers

- Ophthalmology, e.g. for contact lenses with high dioptric power
- imaging applications, e.g. objectives for endoscopes

4.1 Coupling of Optical Fibers

GRIN Lenses are primarily used in coupling of optical fibers minimizing coupling losses. Desirable length of material is used mainly radial index glass which is desirable in this case. Light passing through bend adjust accordingly to the optical wire diameter ensuring smooth transition. There are two main categories for r-GRIN material applications. The initial applications are for sharpening. Assuming that GRIN rod lenses of a currently, optical telephony, signal processing, and micro-optical devices such as an optical fiber have small diameter and at surfaces Disk [7±9]. Choosing the gradient index profile correctly will ensure that the propagation along the rod axis, the course of the light ray fluctuates sinusoidally. The use of such lenses Focused or divergent light that has been collimated, such as concave lens Therefore, these lenses are helpful for collimating or focusing light. Emanating from a source with a small diameter, such as a laser or optical fiber using a diode to optically couple a light source to an optical fiber, as well as coupling.

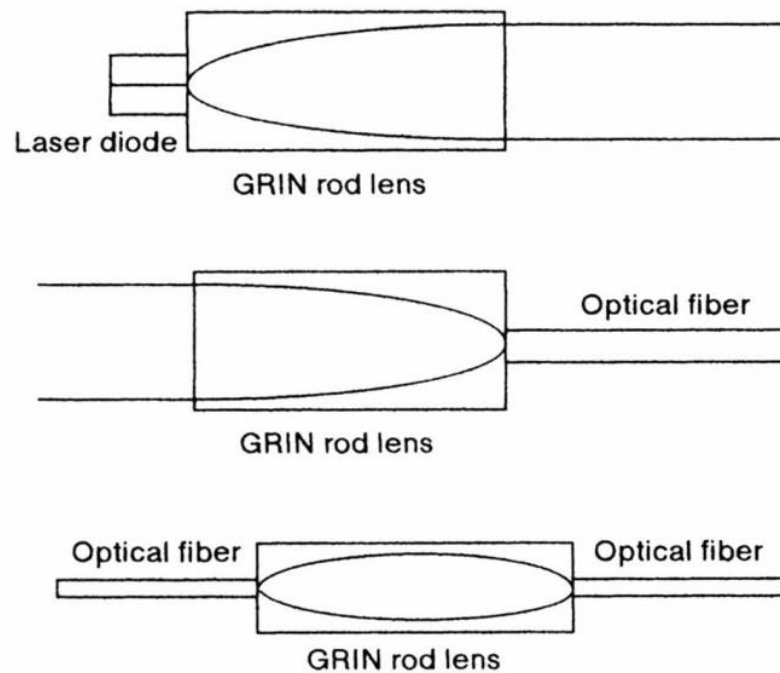


Fig 2 Schematic Cross Section of coupler and collimator using GRIN rod lens with small lens and flat Surface [5]

Gradient optics can only promise an advantage over conventional optics because the ability to produce index of refraction gradients in large optical components is not available.[4]

4.2 Endoscopic Application

The GRIN lenses which are small are an excellent alternative for microscopic proceedings. With High-Proximal lenses these apparatus can be further extenuated.

4.3 Correction Plate

Optics are used more and more by contemporary military systems to carry out target acquisition tasks like missile guiding. The optimum optical design for the target acquisition mission and the best aerodynamic window must be traded off in the design of the missile sensor and its aerodynamic window performance in aerodynamics. The sensors used nowadays only perform basic optical operations. The long-term objective is to be able to use imaging optics to offer some level of autonomy or at the very a minimum increase in accuracy.

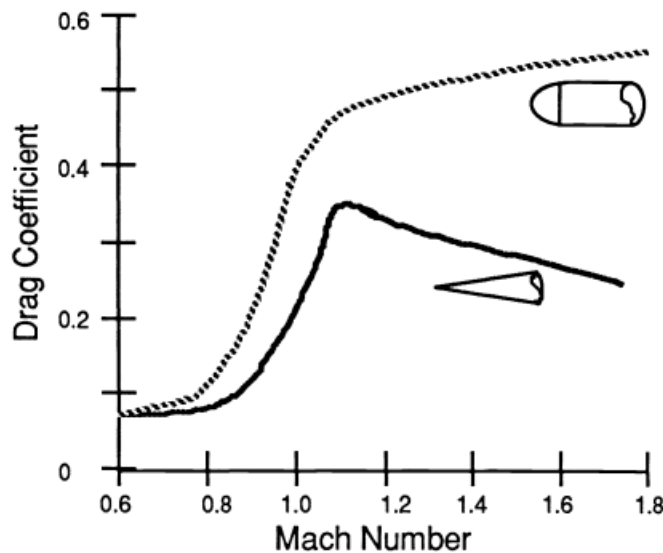


Fig 3 The aerodynamic penalty of using a hemispherical sensor window. The hemispherical window is selected to reduce problems in optical aberrations[4]

5. RESULT AND CONCLUSION

From the evaluated data it is clear that the use case of gradient index glass exceeds far greater than what is being used. With the correct manufacturing methods we can achieve exponential use of this material.

The current analysis suggests that it will take some time for it to attain mass manufacturing and needs a lot of scientific research before being commercialized. A typical glass generally cost the buyer around \$65.68 Dollars which can be reduced massively. The materials being used does not support a large scale manufacturing and thus increases the producing capacity of the glass.

Gradient-index materials are now widely regarded as giving designers of optical systems various extra design freedoms that are beneficial to the lens system. Using additional criteria, such as a designer, curvature, and thickness, modifies how well the lens's index of refraction works.

In the past, a polynomial diagram was used to illustrate how the lens's index of refraction changed during the design process. This mathematical example was useful since it made it simple to calculate the systems first- and third-order attributes without regard to manufacturing conditions. Consequently, building a precise index proved to be difficult an illustration that was used to construct a profile.

In the past, the gradient-index design-to-manufacture process was time-consuming and iterative. After conducting numerous experiments, the manufacturer was able to produce a profile that matched the one the designer requested. The discrepancy between the two profiles decreased optical performance in the majority of systems, necessitating a reoptimization of the lens system. The new design often required a slightly different profile, and the entire process had to be repeated numerous times over the period of several months. From a manufacturing standpoint, it would be preferable to design the lens using the actual fabrication parameters, such as time, temperature, and salt/glass mix.

Gradient-index fabrication by ion exchange is mathematically modeled using Fickian diffusion theory. The model is then tested against experimental data for a variety of glass compositions, diffusion times, and temperatures in both axial and radial sample geometries. Finally, the model is integrated with a lens design program to enable diffusion parameter optimization.

This provides the best output for the glasses end output. This has been the finding of this research on the new age material.

CONCLUSION

Gradient index glass has a high scope be it commercially available. With cheap and easy manufacturing techniques instruments such as endoscopy machines can be made cheap and highly efficient with iterations. High speed communications will particularly spiral with increase in bandwidth capacity

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